INCH-POUND
MIL-I-3190/8A
20 August 1992

MILITARY SPECIFICATION SHEET

# INSULATION SLEEVING, ELECTRICAL, FLEXIBLE, SILICONE COATED, CLASS 220°C, TYPE D, CATEGORY c

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-3190

The sleeving shall be of the following classification:

Class 220°C, Type D, Category c

REQUIREMENTS:

The insulation sleeving shall be as specified in tables I, II, and III.

TABLE I. Performance requirements.

| Property | Conditioning prior to test | Test | Unit | Value required |
|---|---|---|---|---|
| Dielectric breakdown: | | | | |
| Straight | 96/23/50 | 4.7.2.2 | Volts (minimum average) | 8,000 |
| Straight | 96/23/50 | 4.7.2.2 | Volts (minimum individual) | 6,000 |
| Straight | 96/23/96 | 4.7.2.2 | Percent of average value obtained on test after condition 96/23/50 (minimum) | 80 |
| Cold brittleness | 96/23/50 | 4.7.4 | Degrees Celsius (maximum) | minus 70 |
| Flammability | 96/23/50 | 4.7.5.1 Method B | Seconds to burn 1" for each specimen minimum | 45 |

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TABLE I. Performance requirements - Continued.

| Property | Conditioning prior to test | Test | Unit | Value required |
|---|---|---|---|---|
| Push-back after heat aging: | 168/250 | 4.7.8 | Observation | No cracks or rupture in the silver on two or more specimens |
| Visual | | | | |
| Dielectric breakdown: Straight | 168/250 | 4.7.8 | Volts (minimum average) | 6,000 |
| Thermal endurance | see 4.7.9 | 4.7.9 | Temp index for (C) at 15000 hrs minimum | 220 |
| Thermal stability | 96/250 | 4.7.10 | Volts (minimum average) | 5,500 |
| Hydrolytic stability | 336 hours (14 days) over water at 70°C | 4.7.7 | Volts (minimum average) and observation | 5000 No disintegration reversion or cracks |



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TABLE II. Qualification inspection requirements.

| Examination or test | Requirement | Test | Minimum number of tests for each size and for each condition as specified in table I |
|---|---|---|---|
| Visual examination | 3.3, 3.6 | 4.6 | 10 |
| Dimensions | 3.4 | 4.7.1 | 10 |
| Dielectric breakdown: Straight - 96/23/50 and 96/23/96 | Table I | 4.7.2.2 | 10 |
| Cold brittleness | Table I | 4.7.4 | As specified in 4.7.4 (10) |
| Flammability | Table I | 4.7.5.2 Method B | 5 |
| Push-back after heat aging | Table I | 4.7.8 | 3 |
| Thermal endurance | Table I | 4.7.9 | See 4.7.9 |
| Hydrolytic stability | Table I | 4.7.7 | 5 |

TABLE III. Quality conformance inspection requirements.

| Examination or test | Requirement | Test | Conditioning prior to test | Minimum number of tests per sample unit |
|---|---|---|---|---|
| Visual examination | 3.3, 3.6 | 4.6 | 96/23/50 | As required per sampling |
| Dimensions | 3.4 | 4.7.1 | 96/23/50 | As required per sampling |
| Dielectric breakdown: Straight | Table I | 4.7.2.2 | 96/23/50<br>96/23/96 | 10<br>10 |
| Flammability | Table I | 4.7.5.2 Method B | 96/23/50 | 5 |
| Push-back after heat aging | Table I | 4.7.8 | 168/250 | 3 |
| Thermal stability | Table I | 4.7.10 | 96/250 | 5 |



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Intended use: The temperature index for this class of sleeving at 15,000 hours is 220°C. The sleeving covered by this specification is intended for applications in electrical equipment requiring primary insulation having good low temperature flexibility, high moisture and humidity resistance, excellent mechanical protection and abrasion resistance.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - SH
Air Force - 11

Review activities:
Army - AR, AV, EA, MI
Navy - EC
Air Force - 85, 99, 80

User activities:
Army - ME
Navy - MC, OS

Preparing activity:
Navy - SH
(Project 5970-1106)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-I-3190/8A | 920820 |

| 3. DOCUMENT TITLE | CATEGORY |
|---|---|
| INSULATION SLEEVING, ELECTRICAL, FLEXIBLE, SILICONE COATED, CLASS 220°C, TYPE D, | C |

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
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| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

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| a. NAME TECHNICAL POINT OF CONTACT:<br>MR. CHIEN Y. LU, SEA 05E | b. TELEPHONE (Include Area Code)<br>(1) Commercial: 703-602-3124<br>(2) AUTOVON: 332-3124 |
| c. ADDRESS (Include Zip Code)<br>COMMANDER, NAVAL SEA SYSTEMS COMMAND 05Q42<br>2531 NATIONAL CENTER BLDG 3<br>WASHINGTON, DC 20362-5160 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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